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G H Raisoni International Skill Tech University, Pune

T.Y B. Tech (TERM-V)

CAE I- Summer 2026 (2024 Pattern)

Design and Analysis Algorithm (25BTAIPCL502)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed. v) Assume suitable data, if necessary.

Q. No	*Question*	Marks	CO
Q 1. (a)	Define an algorithm and explain its characteristics with suitable examples.	[5]	1
(b)	Describe amortized analysis technique with suitable example.	[5]	1
	OR		
Q 2. (a)	Explain the significance of asymptotic notations in analysis of algorithms.	[5]	1
(b)	Explain the master's theorem for solving recurrence relations in detail.	[5]	1
Q 3. (a)	Apply the Binary Search algorithm to search a given element in a sorted array. Show all intermediate steps. [2, 5, 8, 12, 16, 23, 38, 56, 72] key = 23	[5]	2
(b)	Explain the Divide and Conquer strategy using control abstraction and appropriate example.	[5]	2
	OR		
Q 4. (a)	Apply Strassen's Matrix Multiplication algorithm to find the resulting product matrix C= AX B given $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix}$	[5]	2
(b)	Design an algorithm using the Divide and Conquer approach to find the maximum and minimum elements in an array.	[5]	2